

Week 4

Conditional Expectation and the Best Linear Predictor

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Speaker notes

No notes on this slide.

Speaker notes

No notes on this slide.

Motivating Examples

Motivating Examples

Given a random variable X and another random variable

Y , you often want to make predictions about Y that are *as good as possible*. - Given some budget, what will an organization's head count be next year? - Given a citizen's age and income, how likely are they to vote? - Given a set of sounds picked up by a microphone, what are the chances that someone in the room has fallen down (Sound Flux, 2019)? - Given a set of voxels, does an image contain a malignant tumor?

Speaker notes

- This is the goal of a
- As you may have guessed, we're not just going to head straight to making predictions. We're going to build on top of things that you already know from the last two weeks, and we're going to derive new tools to use that we can

Plan for the Week

Two sections:

1. Conditional expectations
2. Best predictors and best linear predictors

Speaker notes

- First, we'll practice working with covariance and correlation - these are the most foundational math tools that we'll need to work with joint distributions.
- Next, we'll study conditional expectation - this is an important way of summarizing the information in a joint distribution. We can ask, what is the shape of the data, provided that
- Finally, we'll figure out how to make predictions for one random variable, given another - a very important task for data scientists.

Plan for the Week (cont.)

At the end of this week, you will be able to:

- Derive statements describing relationships among random variables
- Understand how to divide variance into a part that is explained and an error component that you have not explained
- Understand how a best linear predictor uses information from a set of random variables to help us predict the value of an outcome

Conditional Expectation

Introduction to Conditional Expectation

Introduction to Conditional Expectation, Part I

To this point, we have characterized random variables with three concepts:

1. The Expected Value , $\mathbb{E}[X]$
2. The Variance , $\mathbb{V}[X]$
3. The Covariance , $\text{cov}[X, Y]$

But, when there is dependence between two random variables X and

Y , then if we know something about X , we might be able to make a new contextualized statement about the *conditional distribution* .

- We've seen how to use expectation and var to summarize a distribution, but we've also seen how observing one variable lets us define a conditional distribution on another variable. Now it's time to put those two concepts together.
- We've got the Frostee. We've got the fries. Let's do this.

Introduction to Conditional Expectation, Part II

We began the week with the questions:

- Given some budget, what will an organization's head count be next year?
- Given some attributes about a citizen, how likely are they to vote?
- Given a set of sounds picked up by a microphone, what are the chances that someone in the room has fallen down (Sound Flux, 2019)?
- Given a set of voxels, does this image contain a malignant tumor?

Introduction to Conditional Expectation, Part III

But, those were all conditional statements!

Rather than saying, “On average, 78 % of the electorate votes in any given election,” we can instead say:

- “Among people 29 or younger, 41 % voted in the last presidential election”
- “Among people who *have* voted before, 90 % will vote in any given election”

This is our goal in data science! Use information to make more informed,

better statements.

Speaker notes

- Conditional statements let us update general knowledge to make more specific,
- It is way lower in off-cycle elections – about 30
- Because there is a probability distribution across these two mutually exclusive cases, we could actually figure out what the population prevalence of these two types are.

Conditional Expectation Operator

Introduction to Conditional Expectation, Part I

- Recall expected values: $E[Y] = \int y f_Y(y) dy$
- The expected value of Y , $E[Y]$, is the integral of the product of { the value y and the probability density function for that value }
- Expectation is an operator that maps from $\mathbb{R}^n \rightarrow \mathbb{R}$

Introduction to Conditional Expectation, Part II

- The conditional expectation operator of Y given X , $\mathbb{E}[Y|X]$ holds a similar form: $\mathbb{E}[Y|X] = \int y f_{Y|X}(y|x) dy$, $X \in \mathcal{X}$
- Recall also that expectation is an operator.
- So, too, is conditional expectation.

Introduction to Conditional Expectation, Part III

Speaker notes

- Draw a conditional distribution function

Conditional Expectation Demo

Software Demo: Conditional Expectation

Note: This is a lecture + software demo. We're just placing this here for organization.

Speaker notes

- In the current version of the course, in video
- We would like to also show this, with the Plotly demo stored in
- In particular, take a sample, build the ``flat'' density functions. Build a slider that students can push-pull. This will give them a sense for what they are conditioning on, and will lay the groundwork for the kernel density section.

Learnosity: Conditional Expectation

Conditional Expectation Example

The following table shows the joint probability distribution for discrete random variables X and Y .

1. Compute: $\mathbb{E}(Y | X = 1)$
2. Compute: $\mathbb{E}(Y | X = 2)$

Reading: Conditional Expectation Operator

Reading: Conditional Expectation Operator

Read page 67 through the first half of 69 (including Theorem 2.2.14)

Properties of Conditional Operators

Understanding Conditional Operators

Conditional Expectation $\rightarrow \mathbb{E}[\text{Conditional Distribution}]$

Conditional Variance $\rightarrow \mathbb{V}[\text{Conditional Distribution}]$

Speaker notes

- As we start to model relationships, some of our most important building blocks are what we call conditional operators. Two of the most important conditional operators are conditional E and conditional V.
- Do you see the pattern here?
- A conditional operator is an operator applied to a conditional distribution. that's it!
- You take the equation for expectation, and then you plug in the conditional probability instead of the regular probability. same for variance
- When you put it that way, you should recognize that pretty much all the properties we derived for regular expectation and regular variance also hold when we condition. We pretty much only have to adjust the notation a bit.
- So we're not going to prove these results (the proofs are

Formulas for Conditional Variance

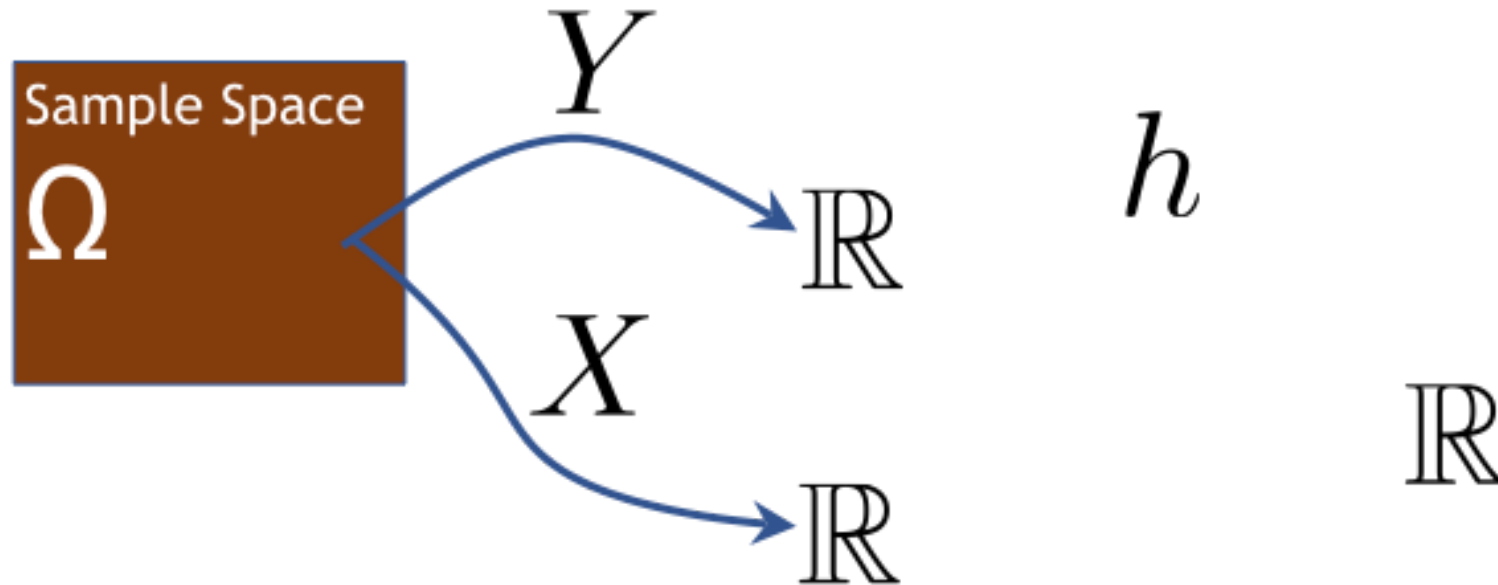
$$V[Y | X = x] = E[(Y - E[Y | X = x])^2 | X = x]$$

$$V[Y | X = x] = E[Y^2 | X = x] - E[Y | X = x]^2$$

Speaker notes

- First, conditional E and conditional V are related just like their unconditional counterparts
- You can write conditional V as an conditional expectation of squared deviations from the conditional mean.
- There's also the more convenient formula, as a conditional expectation of the square, minus the square of the conditional expectation.
- Notice that these equations are the same as before, we're just adding the conditioning to every operator, which means we're putting in a conditional distribution everywhere.

Understanding Conditional Lotus



- Condition on $X = x$.

Speaker notes

- What about that famous rule that we called LOTUS?
- Remember what that was about. if we have a random variable Y , and we pass it into a function h , that creates a new variable and we have a really nice formula for computing its expectation.
- Is there a conditional version of this? Yes! Let's add R.V.
- If we condition on X , that's no problem, we just replace the distribution of Y with the conditional distribution, and plug it into LOTUS.
- Here's an extra little wrinkle though: the theorem you see in the book is a little more powerful than this. It turns out that we can make h depend on both X and Y . that sounds really complicated, but it actually doesn't change much.
- When we condition on x , we treat X as a constant. so h is essentially still just a function of Y .

Theorem: Conditional LOTUS

Given discrete random variables X and Y with joint pmf f , and $h : \mathbb{R}^2 \rightarrow \mathbb{R}$,
 $\mathbb{E}[h(X, Y)|X = x] = \sum_y h(x, y)f_{Y|X}(y|x)$, For all $x \in \text{Supp}[X]$. Given continuous random variables X and Y with joint pdf f ,
 $\mathbb{E}[h(X, Y)|X = x] = \int_{-\infty}^{\infty} h(x, y)f_{Y|X}(y|x)dy$, for all $x \in \text{Supp}[X]$.

Speaker notes

- And here's the full version of the conditional LOTUS.
- The main thing to notice is that the equations look almost the same as before
- we now have the conditional distribution instead of the regular distribution.
- also, h can depend on y , but that doesn't change much because we've conditioned on y , so we can treat it as a constant.

i Theorem: Linearity of Conditional Expectation

Given random variables X and Y and $a, b \in \mathbb{R}$, then for all $x \in \text{Supp}[X]$,
 $\mathbb{E}[aY + b|X = x] = a\mathbb{E}[Y|X = x] + b$ Moreover, given $g, h : \mathbb{R} \rightarrow \mathbb{R}$, then for all
 $x \in \text{Supp}[X]$, $\mathbb{E}[g(X)Y + h(X)|X = x] = g(x)\mathbb{E}[Y|X = x] + h(x)$

Conditional Expectation Example

Simplify: $\mathbb{E}[XY | X = x]$

Demonstration of the Conditional Expectation Function

Demonstration of the CEF

Note: This is a Lecture and Software Demo, we're just placing it here for organization.

Speaker notes

- Show the Plots of Marin Headlands rendered by Plotly.
- Notice that when we're at any particular X value, we have a whole function for the distribution of Y. This is the conditional pdf.
- There is

Reading: Conditional Expectation Function

Reading: Conditional Expectation Function

- Read the second half of page 69, through page 71, stopping before the Law of Iterated Expectations. % \paul{switching to the function view seems like a big % turn, % maybe save that reading till we make the jump?}
- These pages are tough, so we're going to whiteboard theorems 2.2.11 and 2.2.12 on the other side of your reading.
- You can probably skip theorem 2.2.14 without great problem.
- Focus specific attention on definition 2.2.15.

Speaker notes

- We're going to make a couple of notes for you, so grab your book and a pen (if you write in your books)
- Pay particular attention to definition 2.2.15 – this is the pivot from the operator to the function, which has a lot more information. The notation here can be unclear – it isn't your fault.

Learnosity: Write a Conditional Expectation Function

Introduction to the Law of Iterated Expectations

Law of Iterated Expectations

Earlier, we worked with **theorem 1.1.13**, *The law of total probability*, which stated:

Theorem 1.1.13: the law of total probability

If $\{A_1, A_2, A_3, \dots\}$ partition Σ ,
 $B \in \mathcal{S}$, and $P(A_i) > 0$, $\forall i$ then:

$$P(B) = \sum_i P(B|A_i)P(A_i)$$

- Useful because we can break the overall probability into smaller “chunks” whose probabilities might be easier to know.
- The Law of Iterated Expectations is a generalization and reapplication of this principle.

Reading: Law of Iterated Expectations and Total Variance

Reading: Law of Iterated Expectations

Read pages 72, 73, and the top of 74, stopping before theorem 2.2.19.

Lightboard: Law of Iterated Expectations

Law of Iterated Expectations

Speaker notes

- Check out 4.14 law of iterated expectations. It does this in the first half, then there are two examples. I'm cringing now, because I inappropriately assumed a binary gender. But perhaps we can just keep the first part before the examples. and film a separate ipad video with just one or two examples?
- Why is this useful? Well, this says, if you can break down a statement into pieces, you can work them all back together. And, this is a continuous mapping version of the conditional probability statement that we've seen earlier.
- Because we're interested in conditional expectation functions, we're going to use this as a tool quite a bit through the rest of the course.

Law of Total Variance

Splitting Variance

Theorem 2.2.18: the law of total variance

For random variables X and Y :

$$\text{Var}[Y] = \text{E}[\text{Var}[Y|X]] + \text{Var}[\text{E}[Y|X]]$$

Key question: How much of the variance in Y can be explained by X ?

Speaker notes

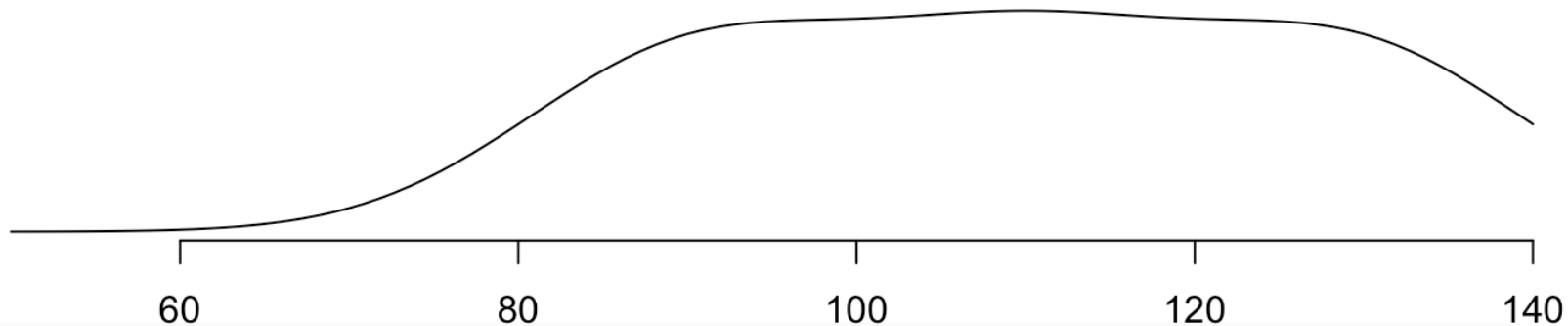
- Remember when we started a few weeks ago from relatively simple axiomatic statements? We've built
- This is the last piece of tooling that we're going to build for this week. In the next sections we're going to apply this tooling against the BP and BLP.
- But, rather happily, this final piece of tooling itself has use – splitting variance!

Splitting Variance: Example

S represents salary.

O represents occupation (1 = data scientist, 2 = data engineer, 3 = data analyst).

Marginal Distribution of Salary



$\mathbb{V}[S] = 366$. How much of this is explained by occupation?

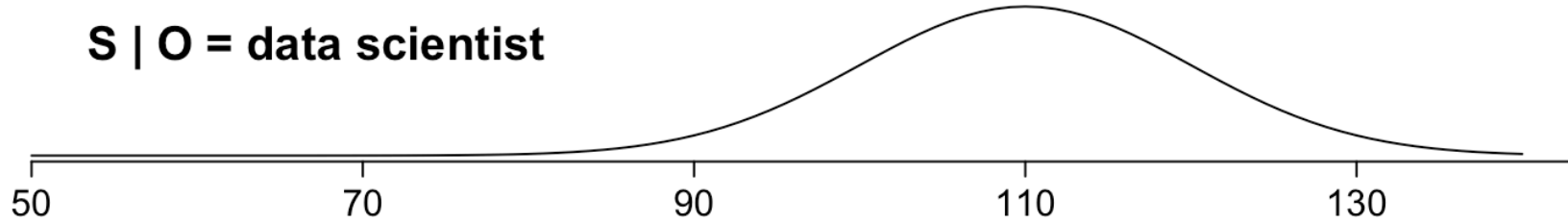
- The law of total variance will work for any random variables, but it's easiest to understand if we condition on a discrete random variable

Salary Conditioned on Occupation

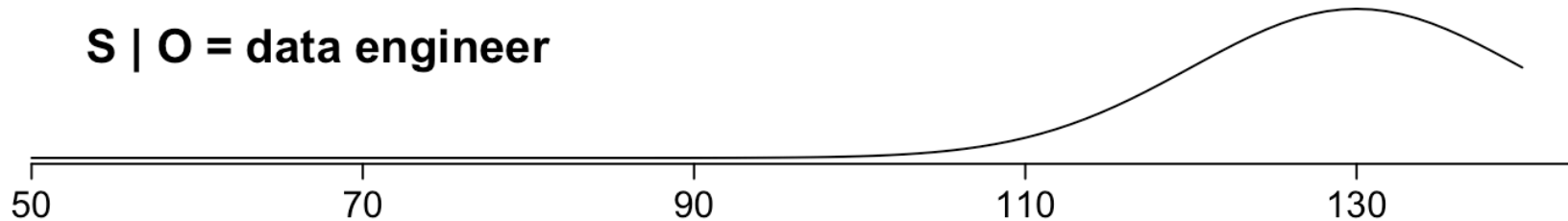
Speaker notes

- Here we see the conditional distributions of S for different occupations. For each one, we can mark
- If these were all equal, then learning occupation doesn't help you make a prediction about salary.
- The more the conditional expectation moves up and down - the more occupation gives us information about the salary
- For each conditional distribution, we can also mark
- The bigger these conditional variances, the more uncertainty there is in salary even after we observe occupation.

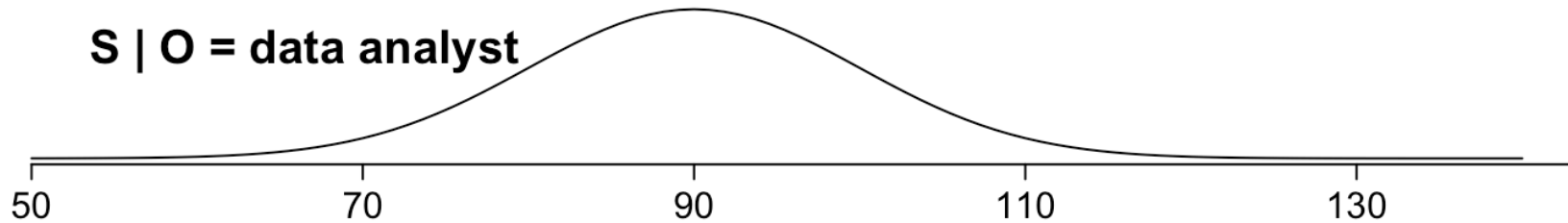
$S \mid O = \text{data scientist}$



$S \mid O = \text{data engineer}$



$S \mid O = \text{data analyst}$



Components of Variance

$V[S]$ is the sum of two components.

1. **Explained variance:** $V[E[S|O]] = 266$ - Measures how much information O gives about S
 - Can also be called systematic variance
2. **Unexplained variance:** $E[V[S|O]] = 100$ - Measures how much extra variation is there in S that we can't explain with O
 - Can also be called error variance

Law of Total Variance Implications

Suppose that you can divide data into groups along (possibly) two dimensions.

1. A dimension that does not explain any variation in outcomes
2. A dimension that explains variation in outcomes

As you're going to see in the next coding exercise, through this identity, if you can produce groups that have different means, you will necessarily produce a smaller $\text{E}[\text{V}[Y|X]]$.

Learnosity: Variance Breakdown

Learnosity: Variance Breakdown

Note: This is a learnosity activity. We're just including it here for organization.

Speaker notes

- Here we will ask students to work the code about temperature and pollution that we've written.

Best Predictors

Deviations From the CEF

More Comments for Alex

Speaker notes

-
- For the proof that CEF minimizes MSE, I think you should first point out that this

Joint Density Function

For this section, we will work first with the same joint distribution function throughout.

$$X \sim U(\min = 0, \max = 10)$$

$$W \sim N(\text{mean} = 0, \text{sd} = 1)$$

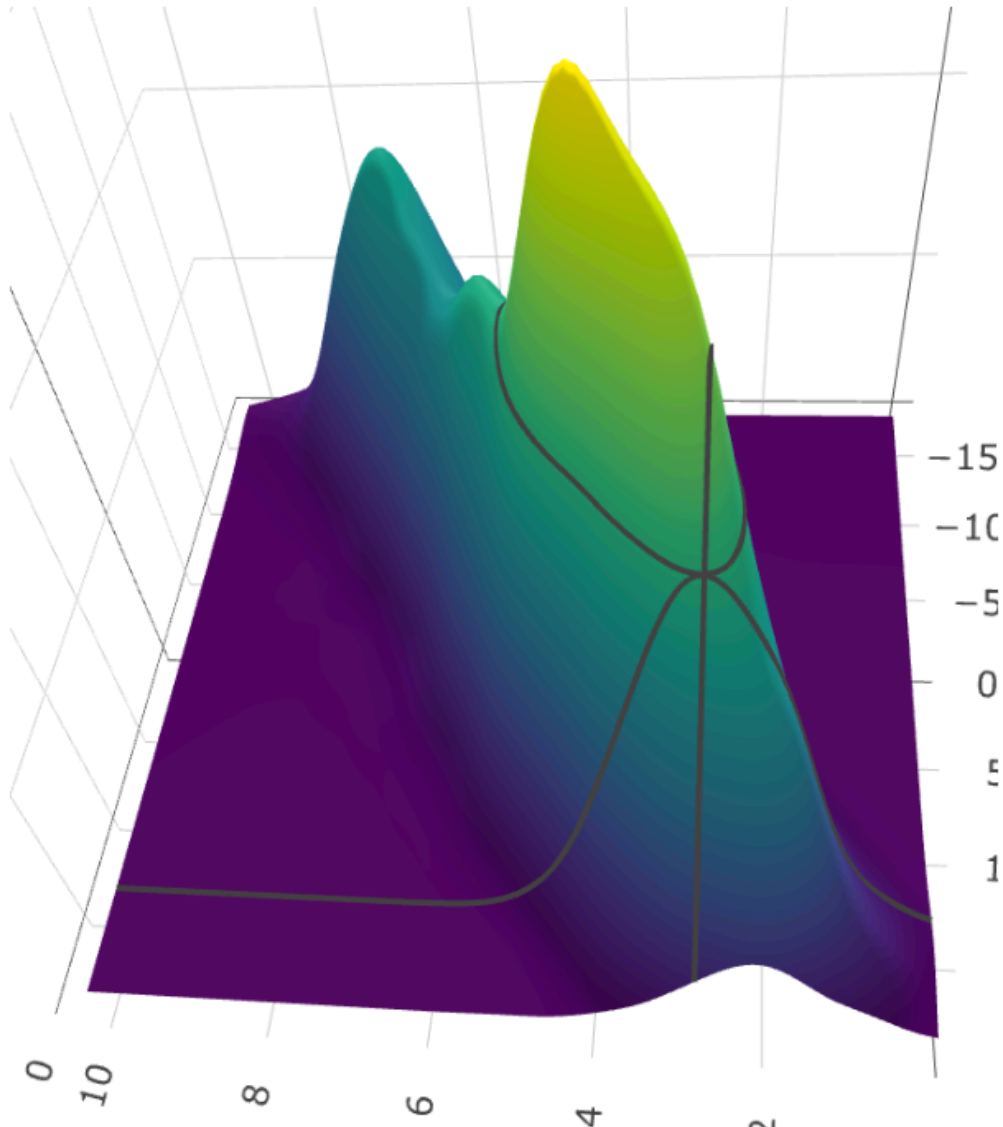
$$Y = 10 - 2 \cdot X + W$$

Where X and W are independent.

Speaker notes

- In this segment of learning, we're going to talk about the Conditional Expectation Function as a quantity of interest.
- We'll begin by defining a quantity
- This error will be presented here, and we'll prove several features about
- Once again, the distinction between an operator and a function is going to be important to keep in mind.
- Notice that here we're providing you the ``diety-eye'' view. We don't get issued this, and we don't know how to estimate these functional forms yet. That will be the focus of the next section.

Joint Density Function (cont.)



Speaker notes

- I like to think about this in terms of a probability landscape – this is a simple one that has a single most probable set of outcomes. In general, this is not the case.
- Callout to the workbook that makes this plot.
- Ask students to rotate the image so that they have made the plot and that the x axis ranges from 0 on the left, and 10 on the right; the y axis from -15 on the furthest to 15 on the nearest.

Deviations from the CEF

i Deviations from the CEF

Suppose that $\epsilon = Y - E[Y|X]$ is the distance between my estimate within a grid and the actual value. \note[item]{Notice that Y is the actual continuous elevation that is modeled by the function of X and Y , but that $E[Y|X]$ is the function that maps the CEF at the point $X=x$ — Inside each of the grids, how far, on average, will I be from the true elevation? $E[\epsilon|X] = ?$

- Across the whole ridgeline, how far, on average, will I be from the true elevation?

$$E[\epsilon] = E[E[\epsilon|X]] = ?$$

- Within each of the grids, what features will shape how close you are, on average?

$$V[\epsilon|X] = ?$$

- Across the whole ridgeline, what features will shape how close you are, on average?

$$V[\epsilon] = ?$$

Reading: Deviations from the CEF

Deviations from the CEF

- Begin by reading from where we left off previously on page 74 to the middle of page 76, stopping when the discussion turns to linear restrictions.
- Try to work through, on your own, the proof of why the CEF is the best (minimum MSE) estimator of Y .
- In the proof of theorem 2.2.20, the authors add $\mathbb{E}[\epsilon|X]$ seemingly from nowhere. This is a legal move because in 2.2.19, you have derived that $\mathbb{E}[\epsilon|X] = 0$.

Best Predictors

Predictors are Functions

Definition: Predictor

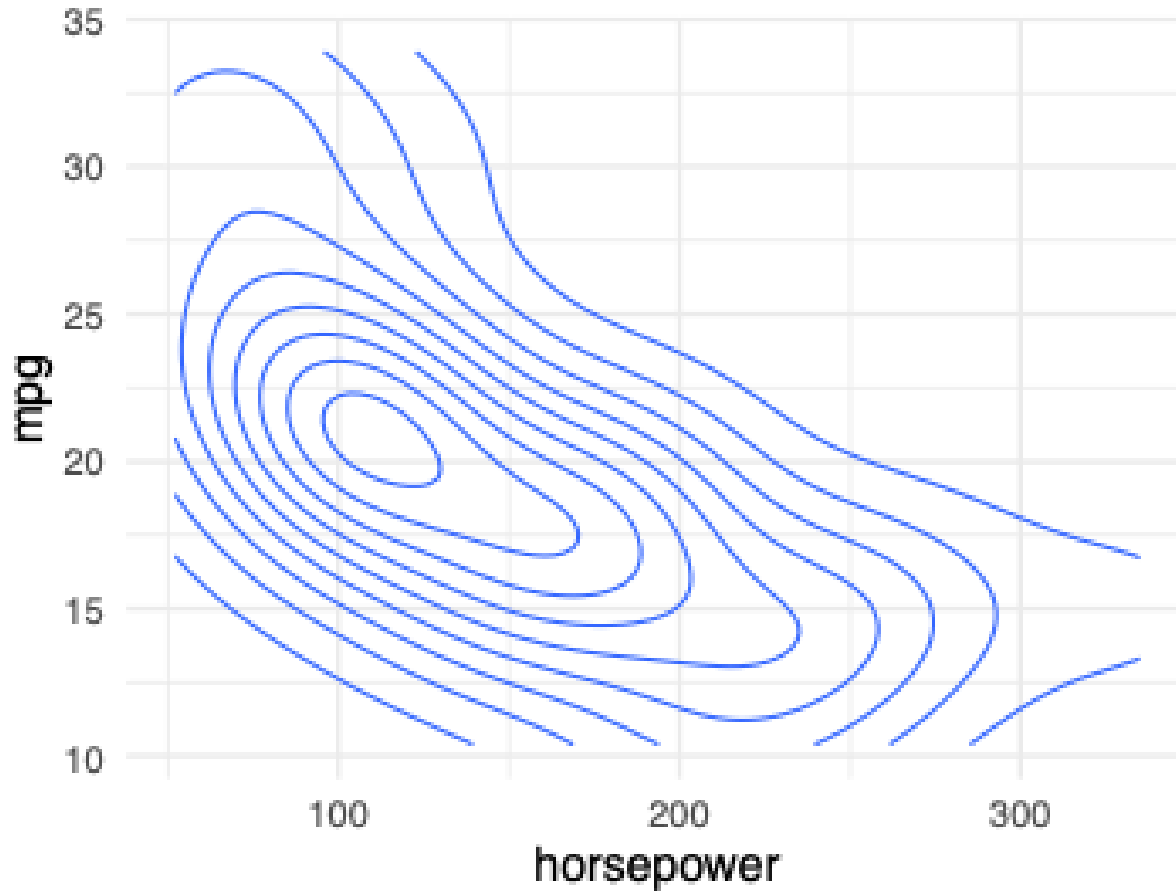
Given random variables X and Y , a predictor for Y is a function, $g : \mathbb{R} \rightarrow \mathbb{R}$, such that $g(X)$ is regarded as a guess for Y .

- Given a value $X = x$, a predictor suggests a single value for Y .
- Define error as $\epsilon = Y - g(X)$.

Speaker notes

- Remember that Y is a random variable, and even if we condition on X , it still have a conditional distribution.
- But we want to represent that distribution with a single number.

Predictors and Errors



Speaker notes

- Let's see an example. Here is a joint distribution, x is hp, y is mpg.
- This is actually an estimated distribution, but let's pretend it's the real one.
- A predictor for mpg is a function. I'll draw one.
- what happens if we take a draw from this distribution?
- Nature reaches in and chooses a point.
- Here is the predicted Y , and this distance is the error.
- The predictor I drew is not a very good predictor. what makes a good predictor?
- We want the error to be small, but how do we do that?

Choosing a Good Predictor

Idea: Minimize mean squared error, $\mathbb{E}[\epsilon^2]$.

Speaker notes

- remember when we had one random variable, we discovered that the mean was the value that minimized expected squared error.
- We can do the same thing here!
- By minimizing expected squared error, we'll be generalizing the mean, and we should end up with a function that stays close to the high probability areas.

Importance of the CEF

Theorem: The CEF Minimizes MSE

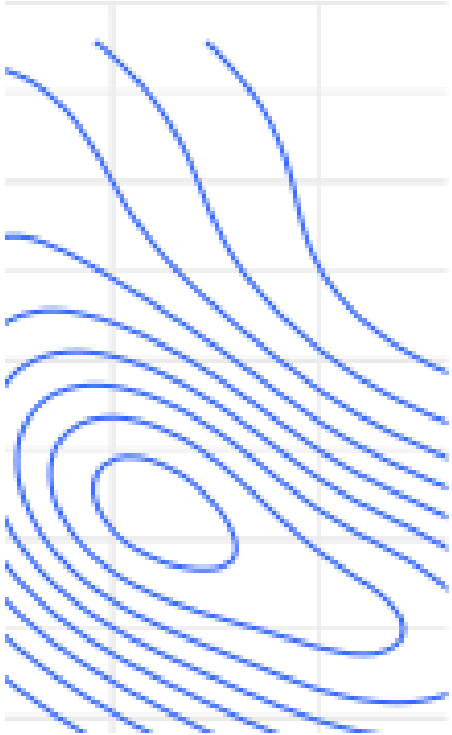
Given random variables X and Y , the CEF $\mathbb{E}[Y|X]$ has the smallest MSE out of all predictors of Y .

Speaker notes

- So what do we get if we minimize MSE? The answer is exactly the CEF!
- If you care about MSE, and you want to generate predictions of
- This is a really important result. It tells us what the maximum possible performance is for any predictor.
- And it helps us to view the CEF from an optimization perspective. That's a viewpoint that really important in both classical statistics and machine learning.

Lightboard: The CEF Minimizes MSE

The CEF minimizes MSE

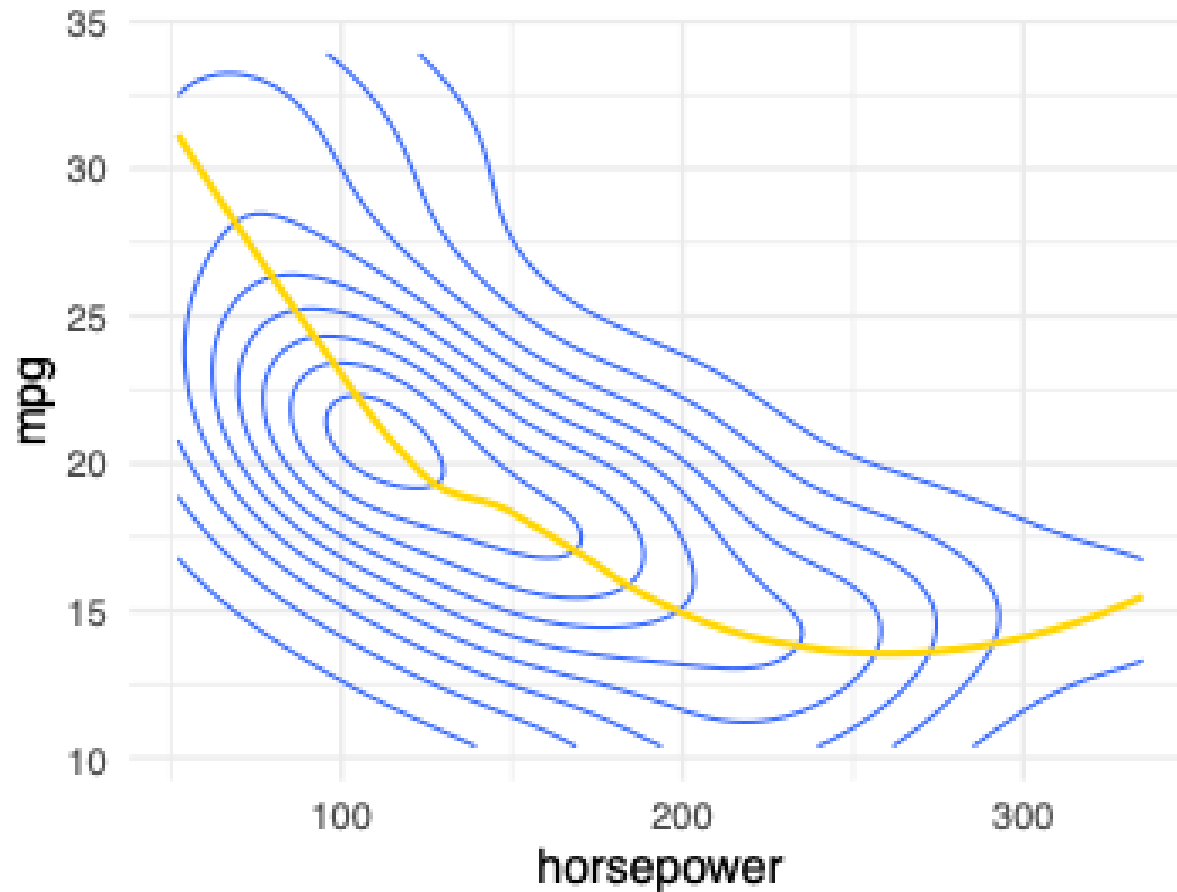


Speaker notes

- draw the CEF
- Let's let
- We want to show that any distance
- write the error:
- Look at the conditional expectation:
- We know
- We can use law of iterated expectations on MSE:

Best Linear Predictors

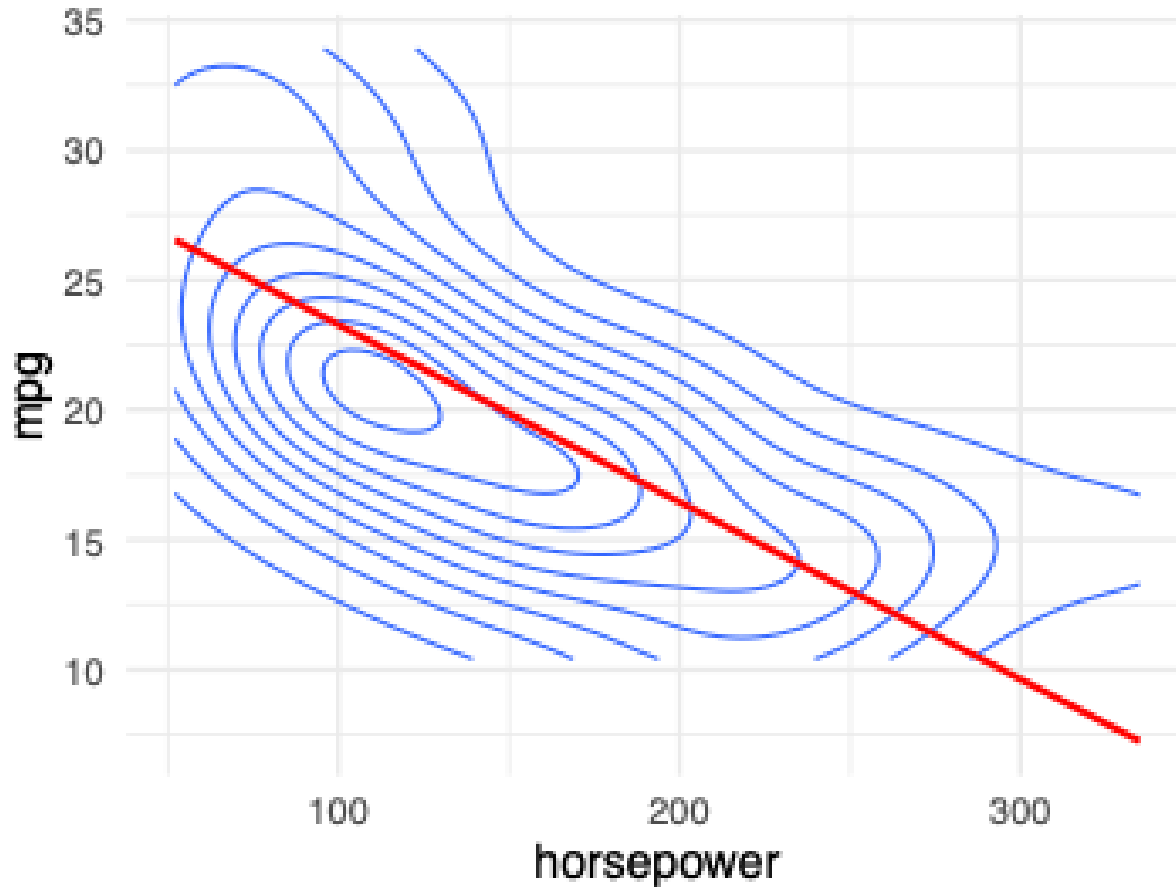
Complexity of the CEF



Speaker notes

- Think about the CEF as a model for some random variable - mpg in this case.
- It's a good model in one sense - it minimizes MSE
- But the CEF can be quite complicated. It takes an infinite number of numbers to specify the CEF.
- If you had all that information, how would you use it?
- Can you reason about every detail in this curve?
- Can you communicate what you see? Can you compare your results against another researcher's?
- The fact is, even if we could estimate the CEF (which takes a LOT of data), we're not well equipped to analyze it directly.
- So you actually don't find too many analyses that estimate the CEF.

Linearizing the Predictor



Speaker notes

- This tells us that we need a simpler model. The simplest possible model that captures a relationship: a line.
- This is what we call a linear predictor.
- There are some disadvantages to using a linear predictor.
- The joint distribution may not look especially linear. We're covering up a lot of complexity.
- Except for very artificial situations, a linear predictor will have worse MSE than the CEF.
- But there are some major advantages.
- You only need a few numbers to write down a linear predictor - the slope is most important.
- The slope is easy to communicate, you can compare studies against each other.
- You have a single number that captures an overall relationship.

Choosing a Linear Predictor

Definition: The Best Linear Predictor (BLP)

Given random variables X and Y , the *best linear predictor (BLP)* for Y is the function $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(X) = \alpha + \beta X$, with coefficients given by,

$$\operatorname{argmin}_{\alpha, \beta} \mathbb{E} \left[\left(Y - (\alpha + \beta X) \right)^2 \right]$$

Speaker notes

- How do you choose a linear predictor?
- There are different strategies, so it depends on your goals
- But the most common idea is to continue our previous strategy, make MSE as small as possible.
- Read definition

Solving for the BLP

Theorem: The BLP Solution

Given random variables X and Y , the best linear predictor for Y is given by $g(X) = \alpha + \beta X$, where,

$$\beta = \frac{\text{cov}[X,Y]}{\text{V}[X]}, \quad \alpha = \text{E}[Y] - \frac{\text{cov}[X,Y]}{\text{V}[X]} \text{E}[X]$$

Speaker notes

- Here's another reason to use the Best Linear Predictor.
- You can solve for the coefficients that minimize MSE, and they have a really elegant solution.
- Remember this form for the slope: it's a covariance over a variance.

Reading: The Best Linear Predictor

Reading Assignment: Linear Approximation

Read to the middle of page 79, stopping before you get to example 2.2.23.

Speaker notes

- This is why we have avoided parameterizing the random variables

Moment Conditions

Moment Conditions

The BLP is given by

$$\operatorname{argmin}_{\alpha, \beta} \mathbf{E} \left[\left(Y - (\alpha + \beta X) \right)^2 \right]$$

Moment Conditions - Solution

The BLP is given by

$$\begin{aligned} \operatorname{argmin}_{\alpha, \beta} \mathbb{E} \left[\left(Y - (\alpha + \beta X) \right)^2 \right] \\ 0 = \frac{\partial \mathbb{E}[\epsilon^2]}{\partial \alpha} &= \mathbb{E} \left[\frac{\partial \epsilon^2}{\partial \alpha} \right] = \mathbb{E} \left[2\epsilon \frac{\partial \epsilon}{\partial \alpha} \right] = -2\mathbb{E}[\epsilon] \\ 0 = \frac{\partial \mathbb{E}[\epsilon^2]}{\partial \beta} &= \mathbb{E} \left[\frac{\partial \epsilon^2}{\partial \beta} \right] = \mathbb{E} \left[2\epsilon \frac{\partial \epsilon}{\partial \beta} \right] = -2\mathbb{E}[\epsilon X] \end{aligned}$$

Version 1:

1. $\mathbb{E}[\epsilon] = 0$
2. $\mathbb{E}[\epsilon X] = 0$

Version 2:

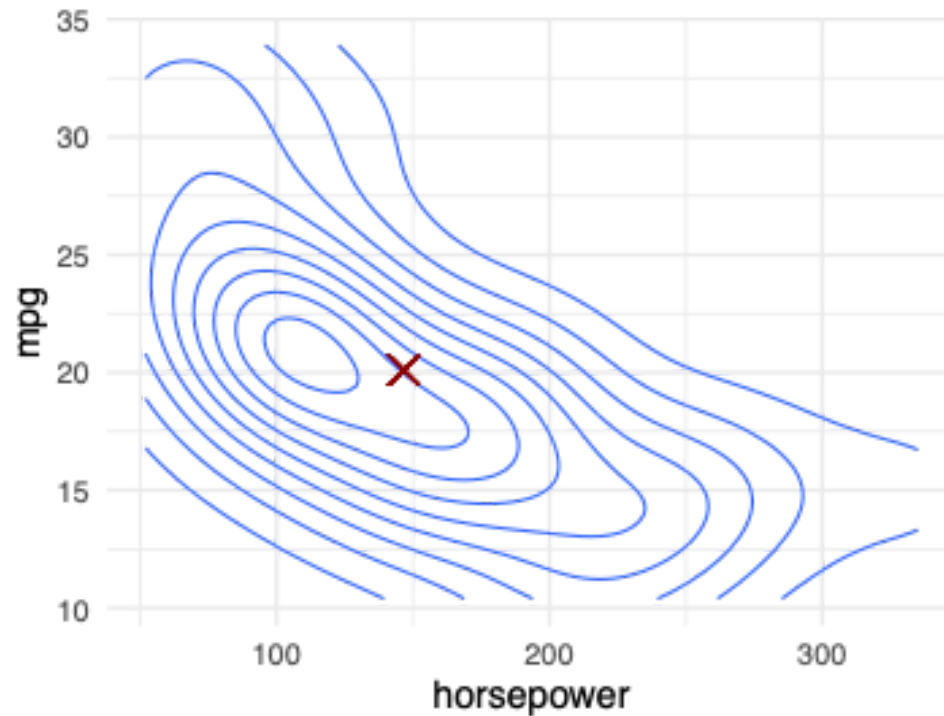
1. $\mathbb{E}[\epsilon] = 0$

Speaker notes

- If you put in different values of
- How do you minimize a function? you can use calculus, use your first order conditions.
- This next step is a little confusing. it turns out that we can move the expectation outside of the derivative. that's because the expectation is a sum or an integral, so it passes through derivatives.
- What's happening here? first, nature reaches in and chooses a data point, which defines an error. then if you move
- It turn out that not only does the BLP fulfill the moment conditions, it's the ONLY line that does. I'm not going to show it, but you can check all the derivatives to see that the MSE has one minimum.

Lightboard: Understanding Moment Conditions

Understanding Moment Conditions



1. $0 = E[\epsilon]$

2. $0 = \text{cov}[\epsilon, X]$

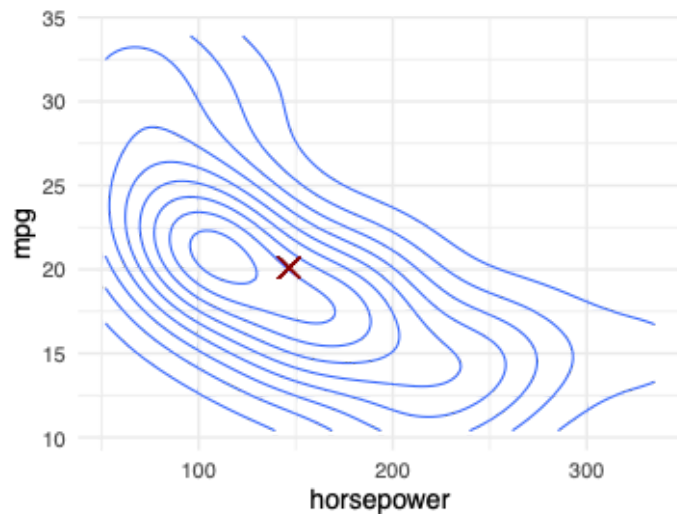
Speaker notes

- Here is a way to think about the moment conditions that I find useful
- Say you want to choose a linear predictor. at first, you're considering every possible line through this plane.
- Now impose the first moment condition. this tells us our line passes through this point. but that's still many different lines. how do we choose one?
- Let's draw one and see what happens. For this line, when X is high, the error tends to be negative. when x is low, the error tends to be positive. so the cov is negative.
- $\text{cov}[\epsilon, X] < 0 \implies$
- $\text{cov}[\epsilon, X] > 0 \implies$

Understanding Moment Conditions - Solution

$$1. 0 = \mathbb{E}[\epsilon] = \mathbb{E}[Y - (\alpha + \beta X)] \Rightarrow \mathbb{E}[Y] = \alpha + \beta \mathbb{E}[X]$$

$$2. 0 = \text{cov}[\epsilon, X]$$



Speaker notes

- Here is a way to think about the moment conditions that I find useful
- Say you want to choose a linear predictor. at first, you're considering every possible line through this plane.
- Now impose the first moment condition. this tells us our line passes through this point. but that's still many different lines. how do we choose one?
- Let's draw one and see what happens. For this line, when X is high, the error tends to be negative. when x is low, the error tends to be positive. so the cov is negative.
- $\text{cov}[\epsilon, X] < 0 \Rightarrow$
- $\text{cov}[\epsilon, X] > 0 \Rightarrow$

Lightboard: The BLP Solution

The BLP Solution

The BLP is defined by the moment conditions:

$$1. 0 = \mathbb{E}[\epsilon]$$

$$2. 0 = \mathbb{E}[\epsilon X]$$

The BLP Solution - Solution

The BLP is defined by the moment conditions:

$$1. 0 = \mathbb{E}[\epsilon] = \mathbb{E}[Y - (\alpha + \beta X)] = \mathbb{E}[Y] - \alpha - \beta \mathbb{E}[X].$$

$$\alpha = \mathbb{E}[Y] - \beta \mathbb{E}[X].$$

$$2. 0 = \mathbb{E}[\epsilon X] = \mathbb{E}[(Y - (\alpha + \beta X))X] = \mathbb{E}[XY] - \alpha \mathbb{E}[X] - \beta \mathbb{E}[X^2]$$

$$= \mathbb{E}[XY] - (\mathbb{E}[Y] - \beta \mathbb{E}[X]) \mathbb{E}[X] - \beta \mathbb{E}[X^2]$$

$$= \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] + \beta \mathbb{E}[X]^2 - \beta \mathbb{E}[X^2]$$

$$= \text{cov}[X, Y] - \beta \text{V}[X]$$

$$\beta = \frac{\text{cov}[X, Y]}{\text{V}[X]} \quad \alpha = \mathbb{E}[Y] - \frac{\text{cov}[X, Y]}{\text{V}[X]} \mathbb{E}[X]$$

Reading: Multivariate Generalizations

Reading: Multivariate Generalizations

Read section 2.3, Multivariate Generalizations.

Best Linear Predictors in Higher Dimensions

Linear Predictors in Higher Dimensions

Definition: Linear Predictor

Given random variable Y , and random vector $\mathbf{X} = (1, X_1, X_2, X_3, \dots, X_k)$ A *linear predictor* for Y is a function, $g : \mathbb{R}^k \rightarrow \mathbb{R}$ of the form,

$g(x_0, x_1, x_2, \dots, x_k) = \beta_0 x_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$ such that $\hat{Y} = g(1, X_1, X_2, \dots, X_k)$ is regarded as a guess for Y .

Interpreting Model Coefficients

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$
$$\frac{\partial \hat{Y}}{\partial X_i} = \beta_i, \quad \Delta \hat{Y} = \beta_i \cdot \Delta X_i$$

i All else equal

- If X_i changes by ΔX_i , the prediction $\hat{Y}$ changes by $\beta_i \cdot \Delta X_i$, *if the other X 's are held constant*.

Speaker notes

- How do we interpret the betas in a linear predictor?
- This actually applies to any linear predictor, not just the best one.
- To get an idea, try taking the partial derivative of the predicted outcome, with respect to some
- This means that you can think of

Interpreting Model Coefficients - Example

Does this model say peacocks with longer tails fly slower?

[= 4.3 - 1.2 + 0.8]



Speaker notes

- Here's a predictor that we developed to describe the flying speed of male peacocks.
- Notice the coefficient on the tail length variable:
- Does this suggest that peacocks with longer tails are slower flyers?
- The answer is no - we have to remember *ceteris paribus*
- The correct interpretation is that peacocks with 1 unit longer tails, but the same muscle mass, are predicted to fly 1.2 units slower. But what about peacocks with longer tails as a whole? We just don't know. For example, peacocks with longer tails might also have more muscle mass. Maybe they fly faster.

Best Linear Predictors in Higher Dimensions

Definition: Best Linear Predictor

Given random variable Y , and random vector $\mathbf{X} = (1, X_1, X_2, X_3, \dots, X_k)$ The *best linear predictor* for Y is the function, $g : \mathbb{R}^{k+1} \rightarrow \mathbb{R}$,

$g(x_0, x_1, x_2, \dots, x_k) = \beta_0 x_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$, with coefficients given by,

$$\operatorname{argmin}_{\beta_0, \dots, \beta_k} \mathbb{E} \left[\left(Y - (\beta_0 + \beta_1 X_1 + \dots + \beta_k X_k) \right)^2 \right]$$

Moment Conditions in Higher Dimensions

Apply first-order conditions:

$$0 = \frac{\partial \mathbb{E}[\epsilon^2]}{\partial \beta_0} = \mathbb{E} \left[\frac{\partial \epsilon^2}{\partial \beta_0} \right] = \mathbb{E} \left[2\epsilon \frac{\partial \epsilon}{\partial \beta_0} \right] = -2\mathbb{E}[\epsilon]$$

$$\text{For } j > 0, 0 = \frac{\partial \mathbb{E}[\epsilon^2]}{\partial \beta_j} = \mathbb{E} \left[\frac{\partial \epsilon^2}{\partial \beta_j} \right] = \mathbb{E} \left[2\epsilon \frac{\partial \epsilon}{\partial \beta_j} \right] = -2\mathbb{E}[\epsilon X_j]$$

Version 1:

1. $\mathbb{E}[\epsilon] = 0$
2. $\mathbb{E}[\epsilon X_j] = 0$

Version 2:

1. $\mathbb{E}[\epsilon] = 0$

The BLP Solution in Higher Dimensions

$$\beta = \mathbb{E} \left[(\mathbb{E} X^T \mathbb{E} X)^{-1} \right] \mathbb{E} \left[\mathbb{E} X^T Y \right]$$

Speaker notes

- Finally, I just want to mention that there is a closed form solution to the BLP, even in higher dimensions.
- This solution requires some matrix algebra, so we won't get into it now.
- We'll come back later and talk about ordinary least squares regression, which is a technique for estimating the BLP, using a sample of data.
- At that point, we'll spend more time explaining the matrix operations.
- For now, just remember that the BLP is a way to summarize the relationship among variables, and it's the target of interest for a lot of research.

Looking Ahead

- The Best Linear Predictor is also called the Population Regression Function.
- Linear Regression: An algorithm for selecting a linear predictor given a sample of data.
- Ordinary Least Squares Regression: An algorithm for estimating the best linear predictor.